

1.07 Differentiation				
1.07a	Gradients	a)	Understand and be able to use the derivative of $f(x)$ as the gradient of the tangent to the graph of $y = f(x)$ at a general point (x, y) .	MG1
1.07b		b)	Understand and be able to use the gradient of the tangent at a point where $x = a$ as: - the limit of the gradient of a chord as x tends to a - a rate of change of y with respect to x. <i>Learners should be able to use the notation $\frac{dy}{dx}$ to denote the rate of change of y with respect to x.</i> <i>Learners should be able to use the notations $f'(x)$ and $\frac{dy}{dx}$ and recognise their equivalence.</i>	
1.07c		c)	Understand and be able to sketch the gradient function for a given curve.	
1.07d		d)	Understand and be able to find second derivatives.	
1.07f			<i>Learners should be able to use the notations $f''(x)$ and $\frac{d^2y}{dx^2}$ and recognise their equivalence.</i>	
1.07e		e)	Understand and be able to use the second derivative as the rate of change of gradient. <i>e.g. For distinguishing between maximum and minimum points.</i> <i>For the application to points of inflection, see 1.07f.</i>	
		f)	Understand and be able to use the second derivative in connection to convex and concave sections of curves and points of inflection. <i>In particular, learners should know that:</i> <i>if $f''(x) > 0$ on an interval, the function is convex in that interval;</i> <i>if $f''(x) < 0$ on an interval the function is concave in that interval;</i> <i>if $f''(x) = 0$ and the curve changes from concave to convex or vice versa there is a point of inflection.</i>	

OCR Ref.	Subject Content	Stage 1 learners should ...	Stage 2 learners additionally should ...	DfE Ref.
1.07g 1.07h	Differentiation from first principles	g) Be able to show differentiation from first principles for small positive integer powers of x . <i>In particular, learners should be able to use the definition</i> $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ <i>including the notation.</i> <i>[Integer powers greater than 4 are excluded.]</i>	h) Be able to show differentiation from first principles for $\sin x$ and $\cos x$.	MG1
1.07i 1.07j 1.07k 1.07l	Differentiation of standard functions	i) Be able to differentiate x^n , for rational values of n , and related constant multiples, sums and differences.	j) Be able to differentiate e^{kx} and a^{kx} , and related sums, differences and constant multiples. k) Be able to differentiate $\sin kx$, $\cos kx$, $\tan kx$ and related sums, differences and constant multiples. l) Understand and be able to use the derivative of $\ln x$.	MG2
1.07m 1.07p 1.07n 1.07o	Tangents, normals, stationary points, increasing and decreasing functions	m) Be able to apply differentiation to find the gradient at a point on a curve and the equations of tangents and normals to a curve. n) Be able to apply differentiation to find and classify stationary points on a curve as either maxima or minima. <i>Classification may involve use of the second derivative or first derivative or other methods.</i> o) Be able to identify where functions are increasing or decreasing. <i>i.e. To be able to use the sign of $\frac{dy}{dx}$ to determine whether the function is increasing or decreasing.</i>	p) Be able to apply differentiation to find points of inflection on a curve. <i>In particular, learners should know that if a curve has a point of inflection at x then $f''(x) = 0$ and there is a sign change in the second derivative on either side of x; if also $f'(x) = 0$ at that point, then the point of inflection is a stationary point, but if $f'(x) \neq 0$ at that point, then the point of inflection is not a stationary point.</i>	MG3

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1.07q 1.07r	Techniques of differentiation		q) Be able to differentiate using the product rule and the quotient rule. r) Be able to differentiate using the chain rule, including problems involving connected rates of change and inverse functions. <i>In particular, learners should be able to use the following relations:</i> $\frac{dy}{dx} = 1 \div \frac{dx}{dy} \text{ and } \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}.$	MG4
1.07s	Parametric and implicit differentiation		s) Be able to differentiate simple functions and relations defined implicitly or parametrically for the first derivative only. <i>They should be able to find the gradient at a point on a curve and to use this to find the equations of tangents and normals, and to solve associated problems.</i> <i>Includes differentiation of functions defined in terms of a parameter using the chain rule.</i>	MG5
1.07t	Constructing differential equations		t) Be able to construct simple differential equations in pure mathematics and in context (contexts may include kinematics, population growth and modelling the relationship between price and demand).	MG6

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